

FIGURE 15.16 Region of integration for Example 3.

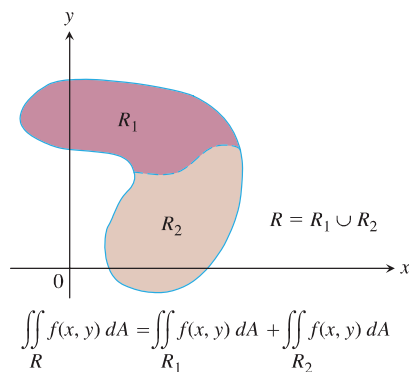


FIGURE 15.17 The Additivity Property for rectangular regions holds for regions bounded by continuous curves.

EXAMPLE 3 Sketch the region of integration for the integral

$$\int_0^2 \int_{x^2}^{2x} (4x + 2) dy dx$$

and write an equivalent integral with the order of integration reversed.

Solution The region of integration is given by the inequalities $x^2 \leq y \leq 2x$ and $0 \leq x \leq 2$. It is therefore the region bounded by the curves $y = x^2$ and $y = 2x$ between $x = 0$ and $x = 2$ (Figure 15.16a).

To find limits for integrating in the reverse order, we imagine a horizontal line passing from left to right through the region. It enters at $x = y/2$ and leaves at $x = \sqrt{y}$. To include all such lines, we let y run from $y = 0$ to $y = 4$ (Figure 15.16b). The integral is

$$\int_0^4 \int_{y/2}^{\sqrt{y}} (4x + 2) dx dy.$$

The common value of these integrals is 8. ■

Properties of Double Integrals

Like single integrals, double integrals of continuous functions have algebraic properties that are useful in computations and applications.

If $f(x, y)$ and $g(x, y)$ are continuous on the bounded region R , then the following properties hold.

1. *Constant Multiple:* $\iint_R cf(x, y) dA = c \iint_R f(x, y) dA$ (any number c)

2. *Sum and Difference:*

$$\iint_R (f(x, y) \pm g(x, y)) dA = \iint_R f(x, y) dA \pm \iint_R g(x, y) dA$$

3. *Domination:*

(a) $\iint_R f(x, y) dA \geq 0$ if $f(x, y) \geq 0$ on R

(b) $\iint_R f(x, y) dA \geq \iint_R g(x, y) dA$ if $f(x, y) \geq g(x, y)$ on R

4. *Additivity:* $\iint_R f(x, y) dA = \iint_{R_1} f(x, y) dA + \iint_{R_2} f(x, y) dA$

if R is the union of two nonoverlapping regions R_1 and R_2

Property 4 assumes that the region of integration R is decomposed into nonoverlapping regions R_1 and R_2 with boundaries consisting of a finite number of line segments or smooth curves. Figure 15.17 illustrates an example of this property.

The idea behind these properties is that integrals behave like sums. If the function $f(x, y)$ is replaced by its constant multiple $cf(x, y)$, then a Riemann sum for f

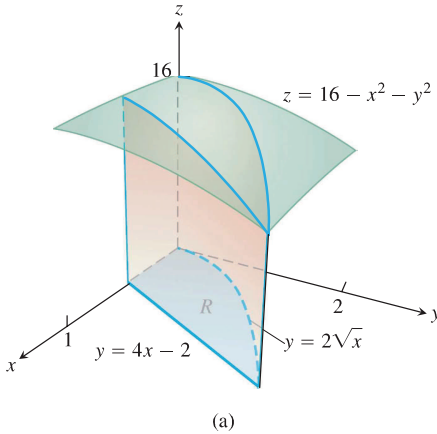
$$S_n = \sum_{k=1}^n f(x_k, y_k) \Delta A_k$$

is replaced by a Riemann sum for cf

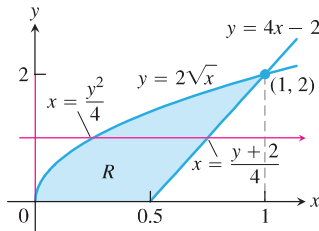
$$\sum_{k=1}^n cf(x_k, y_k) \Delta A_k = c \sum_{k=1}^n f(x_k, y_k) \Delta A_k = cS_n.$$

Taking limits as $n \rightarrow \infty$ shows that $c \lim_{n \rightarrow \infty} S_n = c \iint_R f \, dA$ and $\lim_{n \rightarrow \infty} cS_n = \iint_R cf \, dA$ are equal. It follows that the constant multiple property carries over from sums to double integrals.

The other properties are also easy to verify for Riemann sums, and carry over to double integrals for the same reason. While this discussion gives the idea, an actual proof that these properties hold requires a more careful analysis of how Riemann sums converge.



(a)



(b)

FIGURE 15.18 (a) The solid “wedgelike” region whose volume is found in Example 4. (b) The region of integration R showing the order $dx \, dy$.

EXAMPLE 4 Find the volume of the wedgelike solid that lies beneath the surface $z = 16 - x^2 - y^2$ and above the region R bounded by the curve $y = 2\sqrt{x}$, the line $y = 4x - 2$, and the x -axis.

Solution Figure 15.18a shows the surface and the “wedgelike” solid whose volume we want to calculate. Figure 15.18b shows the region of integration in the xy -plane. If we integrate in the order $dy \, dx$ (first with respect to y and then with respect to x), two integrations will be required because y varies from $y = 0$ to $y = 2\sqrt{x}$ for $0 \leq x \leq 0.5$, and then varies from $y = 4x - 2$ to $y = 2\sqrt{x}$ for $0.5 \leq x \leq 1$. So we choose to integrate in the order $dx \, dy$, which requires only one double integral whose limits of integration are indicated in Figure 15.18b. The volume is then calculated as the iterated integral:

$$\begin{aligned} & \iint_R (16 - x^2 - y^2) \, dA \\ &= \int_0^2 \int_{y^2/4}^{(y+2)/4} (16 - x^2 - y^2) \, dx \, dy \\ &= \int_0^2 \left[16x - \frac{x^3}{3} - xy^2 \right]_{x=y^2/4}^{x=(y+2)/4} dy \\ &= \int_0^2 \left[4(y+2) - \frac{(y+2)^3}{3 \cdot 64} - \frac{(y+2)y^2}{4} - 4y^2 + \frac{y^6}{3 \cdot 64} + \frac{y^4}{4} \right] dy \\ &= \left[\frac{191y}{24} + \frac{63y^2}{32} - \frac{145y^3}{96} - \frac{49y^4}{768} + \frac{y^5}{20} + \frac{y^7}{1344} \right]_0^2 = \frac{20803}{1680} \approx 12.4. \end{aligned}$$

Exercises 15.2

Sketching Regions of Integration

In Exercises 1–8, sketch the described regions of integration.

- $0 \leq x \leq 3, \quad 0 \leq y \leq 2x$
- $-1 \leq x \leq 2, \quad x - 1 \leq y \leq x^2$
- $-2 \leq y \leq 2, \quad y^2 \leq x \leq 4$
- $0 \leq y \leq 1, \quad y \leq x \leq 2y$

- $0 \leq x \leq 1, \quad e^x \leq y \leq e$
- $1 \leq x \leq e^2, \quad 0 \leq y \leq \ln x$
- $0 \leq y \leq 1, \quad 0 \leq x \leq \sin^{-1} y$
- $0 \leq y \leq 8, \quad \frac{1}{4}y \leq x \leq y^{1/3}$